

HCI and HRI Safety Considerations

Concern - Context - Mitigation - Residual Risk

HCI - ED triage desk

1. Alarm fatigue

Context Repeated alerts become background noise.

Mitigation Visual-first alerts; suppress duplicates; audit alert rate.

Residual Thresholds still require local calibration.

2. Legibility under stress

Context Glance-based use under poor lighting and fatigue.

Mitigation High contrast, large text, one primary action.

Residual User testing is still required.

3. Accessibility

Context Colour-only states can exclude or confuse users.

Mitigation Text + icon + colour; keyboard support; plain language.

Residual Unknown needs may emerge during testing.

HRI - Observation Unit kiosk

1. Voice input failure

Context Noise, accents and masks reduce recognition accuracy.

Mitigation Voice is optional; every response is visibly confirmed.

Residual Some users may not use voice at all.

2. Sensor failure

Context Partial or incorrect readings may mislead the model.

Mitigation Block inference, alert nurse and switch to manual mode.

Residual Manual response may be delayed in busy periods.

3. Physical proximity

Context Robot may obstruct, startle or create collision risk.

Mitigation Stationary pilot, clearance zone, emergency stop.

Residual On-site risk assessment remains essential.